

Ferricula v3 Benchmark Plan: What to Run, How, and What Numbers Will Be Credible (September 2026)

Ferricula's white paper will only be credible if every headline number comes from a controlled, paired comparison: memory-on vs memory-off, one component changed at a time, a fixed judge, confidence intervals, and abstentions reported separately. Leaderboard-style single scores on LoCoMo or LongMemEval do not meet that bar in 2026. The must-run core is: LongMemEval-S (cleaned) under the MemDelta protocol, a forgetting/validity track (Memora/FAMA plus STALE), LongBench-Cite plus a self-built exact-quote QA set for the evidence plane, a gate-calibration study with enough labels ($n \geq 500$ per gate per language), and three custom benchmarks: pretraining-controlled persona recall, dream grounding, and lifecycle invariants.

TL;DR

- **Run fewer public benchmarks, more rigorously.** Must-run public sets are LongMemEval-S cleaned (500 Qs), Memora/FAMA and STALE (forgetting/validity), LongBench-Cite (citation precision/recall, includes Chinese), and CIMemories plus a MemoryGraft-style injection test (safety). Treat LoCoMo as a secondary, audit-caveated number and BEAM/LongMemEval-V2/MemoryAgentBench as should-run once the dense arm lands.
- **The Steve Jobs problem needs a design, not a disclaimer.** Split persona questions into public-biography, memory-only episodic (screened by a memory-off closed-book probe), and planted counterfactual items; report the memory-on minus memory-off delta with McNemar's test at ≥ 150 –200 paired items per class. The planned 30 questions give a $\pm \sim 18$ -point 95% interval and cannot support any claim.
- **Current gate numbers cannot be interpreted yet.** On small label sets, ECE's sampling noise is about the size of the 0.05 bar (a published subsampling study reports ECE standard deviation of 0.038 at $n=50$ and 0.0076 at $n=500$). ([arxiv](#)) Rebuild label sets to ≥ 500 items per gate per language with double annotation, use equal-mass/sweep ECE plus Brier and risk-coverage curves, and evaluate the "worth deliberating?" cascade as an accuracy-vs-cost Pareto curve, as JEV-as-a-Judge does.

Key Findings

1. **Protocol matters more than benchmark choice.** MemDelta (arXiv 2606.29914) showed on LongMemEval-S that swapping only the embedding model in an identical verbatim-RAG pipeline moved accuracy from 47.2% to 53.4% (+6.2pp, $p=0.004$), enough to flip a "Mem0 beats RAG by 11pp" conclusion into "Mem0 loses by 1.2pp." It also found model-family reversals: full context beat RAG by 14pp on Gemini 2.5 Flash but lost by ~ 31 pp on Claude Sonnet, where 63% of full-context errors were refusals. A no-memory baseline

scored 2.2%. (arxiv) Ferricula's BM25-only vs BM25+GTR-T5 comparison must therefore be run with everything else frozen.

2. **Vendor leaderboard numbers are not comparable to each other.** Mem0 reports 92.5% LoCoMo / 94.4% LongMemEval / 64.1% BEAM-1M on its own blog; (arxiv) Zep claims 94.7% LoCoMo; Dakera reports 88.2% "with no LLM-assisted reranking"; a technical report claims 481/500 on LongMemEval. (arxiv) (Mem0) Mem0's own peer-comparison paper reported J=66.88 vs Zep 65.99, and Mem0's re-run of Zep produced 58.44% $\pm$ 0.20 against Zep's claimed 84%. (DEV Community) (GitHub) None of these are peer-reviewed, and they use different answer models, judges and reranking.
3. **LoCoMo has documented data and judge problems.** A community audit (Penfield Labs, not peer-reviewed) reports that 6.4% of the answer key is wrong (99 errors) and that the standard GPT-4o-mini judge accepted 62.81% of deliberately wrong but topically adjacent answers, putting the honest ceiling near 93.6%. Common practice also drops the 446 adversarial/refusal questions (22.5%), which removes abstention from the score. (DEV Community) (DEV Community)
4. **Forgetting benchmarks now separate "retained" from "used."** FAMA scores memory-presence criteria and forgetting-absence criteria separately. (arxiv) That fits Ferricula exactly: decayed or superseded content still exists, but must not drive the answer.
5. **History length can invert rankings.** Ground Truth First (arXiv 2607.21962) found that a budgeted curated-map memory leading at three weeks dropped from 96% to 72% recall of evicted content by nine weeks, while a provenance-typed graph rose to 90% (exact $p=0.031$ across six users). (arXiv) Ferricula's 30/365-night lifecycle tests should report tenure curves, not single points.
6. **Crawler-fed memory is an injection surface.** In MemoryGraft (Srivastava & He, arXiv 2512.16962), the attackers "craft malicious entries that masquerade as legitimate successful experiences and inject them into the agent's memory bank via benign-looking content such as README files." Retrieval was union BM25 + embedding. The grafted memories led the agent to adopt "unsafe patterns such as skipping tests or force-pushing code," and this "persists across sessions until the memory is purged." Ferricula's curiosity drive ingests arbitrary web pages into a BM25 index, so this attack applies directly.

Prioritized Benchmark Plan

Priority	Benchmark	Ferricula component	Why
Must-run	LongMemEval-S (cleaned), MemDelta protocol	Experience plane, hybrid recall	Field-standard comparison point; five abilities incl. knowledge-update and abstention
Must-run	Memora + FAMA	Experience plane (decay/release)	Penalises use of invalidated memory; matches "decay changes priority, not content"
Must-run	STALE	Experience plane, drift	Implicit invalidation plus premise resistance
Must-run	LongBench-Cite (+ self-built exact-quote QA)	Evidence plane	Citation recall/precision/F1; has Chinese subsets
Must-run	Gate calibration study (custom, rebuilt label sets)	Gates/judges	Current sets too small to test ECE<0.05
Must-run	Cascade cost-accuracy curve (custom, JEV-style)	"Worth deliberating?" judge	Justifies the escalation design
Must-run	Pretraining-controlled persona recall (custom)	Steve base memory	Core methodological problem
Must-run	Dream/reflection grounding (custom)	Drives, dream channel	Only way to support "dreams are grounded in traces"
Must-run	Lifecycle invariants, 30/365 nights (custom)	Experience plane	Proves no silent deletion/overwrite
Must-run	CIMemories + injection probe via crawler	Safety	Privacy flow + poisoning via curiosity ingestion
Must-run	Deceased-persona guardrail suite (custom, literature-based)	Safety, persona	Real, deceased person simulated
Should-run	LoCoMo (with audit caveats, cat. 5 reported)	Experience plane	Comparability with vendor claims only

Priority	Benchmark	Ferricula component	Why
Should-run	MemoryAgentBench (esp. FactConsolidation)	Experience plane	Counterfactual supersession resists pretraining
Should-run	BEAM (128K/1M tiers)	Evidence + experience	Scale where context-stuffing fails
Should-run	PersistBench, MemSyco-Bench	Safety, gates	Memory-induced sycophancy, cross-domain leakage
Should-run	HaluMem	Experience plane write path	Extraction/update hallucination by stage
Should-run	Drift ablation (custom)	Papañca/valence monitors	Monitors on vs off
Should-run	72-hour soak + cost/day (custom)	Drives	Operational credibility
Should-run	Curiosity "solution-injection" test (custom, after arXiv 2604.17609)	Curiosity drive	Discovery vs use
Nice-to-have	LongMemEval-V2	Agentic memory	Web-agent trajectories; off-profile for a single persona agent
Nice-to-have	NovelQA (leaderboard), NarrativeQA/QASPER	Evidence plane	Long-document QA; contamination caveats
Nice-to-have	ALFWorld / WebShop / τ^2 -bench (JitMem setting)	Task memory	Ferricula is not primarily a task agent
Nice-to-have	Persona consistency (PersonaGym, InCharacter, CharacterEval)	Persona	Style/consistency, not memory
Nice-to-have	Chinese/Pāli retrieval sets	Multilingual	Blocked until tokenizer/embedder support

Details by Component

1. Experience plane: conversational/agent memory benchmarks

LongMemEval (arXiv 2410.10813, ICLR 2025) — must-run.

- *Tests*: information extraction, multi-session reasoning, temporal reasoning, knowledge updates, abstention. (arXiv)
- *Size*: 500 questions: 70 single-session-user, 56 single-session-assistant, 30 preference, 133 multi-session, 78 knowledge-update, 133 temporal. (arxiv) LongMemEval_S has ~115k tokens of history per question (30-40 sessions); LongMemEval_M has ~500 sessions (~1.5M tokens). (arxiv) The authors released a cleaned version in September 2025 to remove history sessions that interfered with answer correctness; use (longmemeval_s_cleaned.json) and say so. (GitHub) Code and data are public on GitHub/Hugging Face.
- *Protocol*: follow MemDelta's per-question loop: reset a fresh store → ingest → build context → answer at temperature 0 → judge with a fixed binary prompt. (arxiv) For Ferricula, this means a benchmark mode with an **empty experience plane** (never the Steve store), with decay/consolidation both on and off as an ablation. Report per-type accuracy, not just overall.
- *Baselines to include (from MemDelta)*: S0 no-memory, random chunks, full context, verbatim BM25 RAG, and Ferricula full pipeline, all with the same answer model and judge. (arxiv)
- *Pitfalls*: judge model choice, answer-model family (rankings reverse across families), embedding choice, and reranking all shift scores by several points; disclose all of them. Wu et al. report "commercial chat assistants and long-context LLMs showing a 30% accuracy drop on memorizing information across sustained interactions."
- *Reference scores*: MemDelta (peer-review status: preprint) reports GPT-4o-mini verbatim RAG 47.2% [42.8, 51.4], full context 49.8%, cloud-embedding RAG 53.4%. (arxiv) Vendor claims run from ~85% up to 94.4% and 481/500; cite these only as vendor-reported.
- *Cost (estimate)*: ingestion into lume/BM25 is local and cheap; answering 500 questions at ~5-10k retrieved tokens each is ~2.5-5M input tokens plus 500 short judge calls, i.e., low tens of US dollars with a mini-class model. MemDelta notes Mem0's write path cost ~\$0.50+ and ~120 minutes per instance with 1,000+ LLM calls. (arxiv) Report Ferricula's write-path cost too, since its gates run locally.

LoCoMo — should-run, with caveats. 50 conversations; 1,540 questions in the four commonly scored categories plus 446 adversarial/refusal questions. (arxiv +2) Report all five categories, publish your judge prompt, and include the audit caveats: 6.4% key errors, and a 62.81% false-accept rate for the standard GPT-4o-mini judge. (DEV Community) ATANT v1.1 (arXiv

2604.10981) also notes that category 4 (42% of items) is scored by substring containment, and that several practitioners could not reproduce published scores (mem0 issues #2800, #3943, #3944). (arxiv) Use LoCoMo only for comparability, never as a headline.

BEAM (arXiv 2510.27246, ICLR 2026) — should-run. 100 conversations, 2,000 validated questions, tiers at 128K/500K/1M/10M tokens, ten memory abilities including contradiction resolution, event ordering and instruction following. (ICLR 2026) (Tiptreesystems) Scoring is rubric/nugget LLM-as-judge (0/0.5/1 per atomic claim, per a third-party implementer's documentation). (Mnemoverse) The authors' LIGHT framework improves 3.5–12.69% over the strongest baselines. (ICLR 2026) BEAM's value for Ferricula is that the 1M/10M tiers cannot be context-stuffed. (Vectorize) Pull (arXiv 2609.14773) reports BEAM-1M F1 0.464 vs 0.299 for truncation on 263 questions, at a mean of 64.5K tokens. (arxiv) Vendor "#1 on BEAM" claims (Hindsight, Mem0 64.1%) are self-reported. (Vectorize) (Mem0) Start with 128K and 1M tiers to control cost.

MemoryAgentBench (arXiv 2507.05257, ICLR 2026) — should-run. Four competencies: accurate retrieval, test-time learning, long-range understanding, selective forgetting. (arxiv) (arxiv) Inputs are chunked (512 or 4,096 tokens) and fed incrementally, "inject once, query multiple times." (GitHub) (Emergent Mind) Its **FactConsolidation** task uses MQuAKE-derived counterfactual edits where the newer fact carries the larger index, so answering from pretraining is penalised. (arXiv) (arXiv) This is useful for Ferricula's knowledge-update story. A secondary source reports HippoRAG-v2 54.0% and BM25 48.0% on single-hop FactConsolidation, a good reference point for a BM25-first system (not re-verified against the original table). (arXiv)

LongMemEval-V2 (arXiv 2605.12493, ICML 2026) — nice-to-have. 451 curated questions over multimodal web-agent trajectories, with histories beyond 100M tokens. (ICML) (arXiv) Five abilities: static state recall, dynamic state tracking, workflow knowledge, environment gotchas, premise awareness. (ResearchGate) Frontier LLMs reach at most 14.1% without trajectory evidence, so it is naturally pretraining-controlled. The best reported system (AgentRunbook-C) reaches 74.9% (Small) and 70.1% (Medium). (ICML) It is off-profile for a single-persona agent without web-agent trajectories; run it only if the curiosity drive produces comparable browsing traces.

HaluMem (arXiv 2511.03506) — should-run. Operation-level evaluation of extraction, updating and QA; HaluMem-Medium and HaluMem-Long have ~15k memory points and ~3.5k questions. (arXiv) It requires APIs for add-dialogue, get-memories and retrieve. (The Moonlight) Ferricula's write path is mostly verbatim, so extraction hallucination should be near zero by construction; demonstrating that is a cheap, differentiating result.

Which benchmarks report cost/tokens: MemDelta (write-path calls, wall-clock, dollars), Pull (token mean per question), LongBench-Cite (GPT-4o evaluation ≈\$4 for correctness and ≈\$25

for citation quality per run), LME-V2 (accuracy-latency trade-off), JEV-as-a-Judge (fee ratio), and vendor posts such as AgentOS ("LongMemEval-S 85.6% at \$0.0090/correct", vendor-reported). (arxiv +3) Ferricula should report \$/correct and tokens/question on every benchmark.

2. Forgetting, updating, staleness, validity

Ferricula's semantics are unusual: decay lowers priority but keeps content, and text leaves only through logged release decisions. The right benchmarks test behavioural forgetting (not using stale or invalid content), not storage deletion.

- **Memora + FAMA (arXiv 2604.20006, ACL Findings 2026) — must-run.** Weeks-to-months conversations; tasks: remembering, reasoning, recommending. $FAMA = \max(0, MPA - \lambda \cdot (1 - FAA))$, where MPA is memory-presence accuracy, FAA is forgetting-absence accuracy, and $\lambda = N_forget / (N_presence + N_forget)$. The paper evaluates four LLMs and six memory agents and finds frequent reuse of invalidated memories. (OpenReview) (arxiv) For Ferricula, report MPA and FAA separately as well as FAMA. FAA measures whether decayed/superseded records stay out of answers even though they remain in storage.
- **STALE (arXiv 2605.06527) — must-run.** 400 expert-validated conflict scenarios, 1,200 queries, >100 topics, contexts up to 150K tokens, (arXiv) three probes: State Resolution, Premise Resistance, Implicit Policy Adaptation. (arXiv) (arXiv) Probe questions follow a "zero leakage" rule. (arXiv) A later paper (arXiv 2609.01852) summarises the best model at 55.2%. (arxiv) A secondary blog reports one frontier model catching stale facts 92% of the time when asked directly but 30% when the question depended on them implicitly (unverified). (DEV Community) Distractor sessions are drawn from LongMemEval, a limitation the authors note. (arXiv)
- **ForgetEval (in arXiv 2606.15903) — should-run as a unit test.** Decomposes forgetting into supersession, decay, amnesia, purge and drift, with deterministic substring scoring and an adapter protocol, so no LLM judge is needed. (arxiv) Map Ferricula's release decisions (forgiveness/archive) to purge/amnesia and its decay to decay. The paper's "axis flip" (MemPalace 0/385 on ForgetEval-Adv vs 60/150 on Memora) shows why you need both a recall-heavy and a forgetting-heavy benchmark. (arXiv)
- **Ground Truth First (arXiv 2607.21962) — design template.** Facts are planted with validity intervals, volatility classes and sent/received trust before any text is rendered; as-of-date question sets at weeks 3/6/9 produce tenure curves; five architectures are compared against a no-memory control with a fixed answerer, a versioned judge and three replicates. A deterministic classifier separates wrong-specific answers from abstentions. (arXiv +2) Borrow this instrument for the 30/365-night lifecycle runs.
- **Memory Trust Gap (arXiv 2609.01852) — design template.** A Benefit suite (unsolvable without memory) and a Safety suite (an authoritative tool holds the current value, so

following stale memory scores *below* no-memory). 300 frozen base scenarios, SHA-256-pinned, with a cluster bootstrap over 33 template families. (arxiv) This is a clean way to show Ferricula's evidence plane overriding stale experience.

- **PersistBench (arXiv 2602.01146, ICML 2026) and MemSyco-Bench (arXiv 2607.01071) — should-run.** PersistBench measures cross-domain leakage, memory-induced sycophancy and beneficial memory use (failure rates; an Inspect-native implementation is recommended). (GitHub) MemSyco-Bench has five tasks (objective fact judgment, contextual scope control, memory-evidence conflict, personalized memory use, valid memory selection), each scored for generation accuracy plus a sycophancy rate or memory-use metric. (arXiv) (arxiv)

3. Evidence plane: faithful, verbatim, cited QA

Findability (recall@k, MRR, byte fidelity) is necessary but not sufficient. The "photographic memory" claim needs three further measurements: (a) **quote exactness**: the answer's quoted span is a byte-exact substring of the cited section; (b) **citation precision/recall**: each statement is supported by its citations, and each citation is needed; (c) **answer correctness conditional on citation**.

- **LongBench-Cite (arXiv 2409.02897) — must-run.** Sentence-level citations over MultiFieldQA-en/zh, HotpotQA, DuReader (Chinese), GovReport and LongBench-Chat. (arxiv) Metrics are citation recall, precision, F1, average citation length, and a correctness ratio (with vs without citations). (arXiv) (arxiv) GPT-4o judge agreement with humans was Cohen's κ 0.593 (recall) and 0.655 (precision). (ResearchGate) By a secondary summary, proprietary LLMs reached only about 0.5 citation F1 on LongBench-Chat and HotpotQA; I have not checked this against the paper's table. The abstract itself says only that the benchmark reveals "considerable room for improvement" and that the authors' LongCite models surpass "advanced proprietary models including GPT-4o." A GPT-4o evaluation run costs roughly \$4 (correctness) plus \$25 (citation). (ACL Anthology) Because Ferricula cites verbatim sections, report citation length too: short, exact citations are its selling point.
- **L-CiteEval (arXiv 2410.02115) — should-run.** Similar citation precision/recall algorithm with entailment checks; useful as a second citation benchmark. (arxiv)
- **ALCE (Gao et al., arXiv 2305.14627) — reference methodology** for citation recall/precision via NLI; LongBench-Cite builds on it. (Not re-verified in this research pass.)
- **NovelQA (arXiv 2403.12766) — nice-to-have.** 2,305 expert questions over 89 novels (one downstream paper counts 88), contexts over 200K tokens, with generative and multiple-choice modes and an evidence-recall evaluation. (arXiv +2) Gold answers are withheld to prevent contamination, so scoring goes through the leaderboard. (arXiv) The

paper includes a **closed-book** setting because the novels are well known: the same pretraining problem Ferricula has with Monte Cristo. (arXiv)

- **QASPER, NarrativeQA — nice-to-have** (arXiv 2105.03011 and 1712.07040; not re-verified here). QASPER gives evidence-annotated QA over NLP papers and fits the "papers" half of the exact-quote claim.

Self-built exact-quote QA (must-run). Your internal Drive research doc describes a "Monte Cristo Arena" in which the full novel is ingested. (mcp Google Drive ...) *The Count of Monte Cristo* is in every LLM's pretraining data, so design the set as follows:

1. ~300 questions whose answers are verbatim spans, stratified by position in the book and by span length.
2. Score **byte-exact span match** against the cited section, plus citation precision/recall.
3. Run **memory-off (closed-book)** on the same questions; report the delta and the fraction of memory-off "correct" quotes that are actually paraphrases.
4. Add a **perturbed-edition control**: ingest a copy with ~50 planted, logged textual alterations (changed names, numbers, sentences). Memory-on must quote the altered text; any quote of the canonical text is a pretraining leak. This is the cleanest test of "verbatim from store, not from weights."
5. Repeat on 2026 papers published after the answer model's training cutoff.

Multilingual.

- Chinese: use LongBench-Cite's MultiFieldQA-zh and DuReader subsets now, and CharacterEval for Chinese persona dialogue. Add a public Chinese retrieval set such as MIRACL-zh or C-MTEB retrieval tasks (not verified in this pass) once the tokenizer stops ASCII-folding.
- Pāli: I found no established Pāli retrieval benchmark. Build a small parallel Pāli-English sutta retrieval set, e.g. from SuttaCentral's openly licensed segment-aligned texts (licence to be confirmed). Measure diacritic-collision rates caused by ASCII folding (ā/a, ṃ/m, ñ/n) as a tokenizer bug metric.
- GTR-T5 is English-only, so report multilingual results per language, arm by arm (BM25-only vs hybrid), and do not average them into an overall figure.

4. Gates/judges: calibration, selective prediction, cascades

Why current ECE results are not interpretable. ECE estimates are biased and high-variance at small n. Roelofs et al. (arXiv 2012.08668) show equal-mass binning has lower bias than equal-width binning, propose the monotonic-sweep estimator ECE_{sweep} , and find estimation bias is worst for *perfectly calibrated* models. Because ECE is non-negative, noise alone pushes it above zero. (arXiv) (Tiptreesystems) A 2026 subsampling experiment (arXiv

2601.07806) reported ECE std of 0.0381 at $n=50$, 0.0256 at $n=100$, 0.0154 at $n=250$ and 0.0076 at $n=500$. (arxiv) With $n \approx 100$, the sampling noise alone is about half the 0.05 bar.

Protocol (must-run):

- **Labels:** ≥ 500 items per gate per language, stratified by class, double-annotated with Cohen's κ reported; adjudicate disagreements; keep a frozen held-out test split separate from any temperature-scaling split. For 6-way recognition tags, report per-class counts; small classes need their own error bars.
- **Metrics:** accuracy/macro-F1; ECE with equal-mass bins (10–15) **and** ECE_sweep, with bootstrap 95% CIs; Brier score (a proper scoring rule that also rewards sharpness); (MetricGate) reliability diagrams; classwise ECE for the 3-way and 6-way gates.
- **Selective prediction:** risk-coverage curves and AURC (Geifman & El-Yaniv's selective classification framework; not re-verified here); report selective risk at fixed coverages (e.g., 80/90/95%) and the abstention rate separately from accuracy.
- **Cascade evaluation (the "worth deliberating?" judge):** follow JEV-as-a-Judge (arXiv 2609.26550). It compared a decision-only judge against sixteen generative and reward-model judges with blinded human adjudication, and found it within three percentage points of the strongest LLM judge at 0.36% of the fee, with the gap concentrated in low-confidence decisions. (Hugging Face) For Ferricula, sweep the escalation threshold τ and plot accuracy vs escalation rate vs \$ and latency, against two anchors: gate-only and always-escalate LLM. Report the fraction of final errors that came from confidently accepted items; that is the cascade's real failure mode.
- **Public reference sets:** for 3-way feeling-tone, a public 3-class sentiment set (e.g., TweetEval sentiment, not verified here) gives an external sanity check, with the caveat that *vedanā* (pleasant/unpleasant/neutral feeling-tone) is not the same construct as sentiment polarity. For leading-question resistance, use STALE's Premise Resistance probe and MemSyco-Bench's sycophancy rate.

5. Grounding of dreams and reflections

Use claim-level decomposition (FActScore, arXiv 2305.14251, not re-verified here): split each dream or reflection into atomic claims and label each claim's provenance: (i) supported by the day's traces; (ii) supported by an entropy-drawn distant memory (logged IDs); (iii) supported by an ingested document; (iv) consistent with public Steve Jobs biography but absent from all supplied inputs, i.e., **pretraining leakage**; (v) unsupported/novel.

- Automate support labels with an NLI/alignment scorer (AlignScore, arXiv 2305.16739; or RAGAS-style faithfulness, arXiv 2309.15217; neither re-verified here), calibrated against a human-labelled audit subset of ≥ 200 claims with κ reported.
- Dreams are *supposed* to recombine, so "unsupported" is not automatically a failure. The white-paper claims should be: (a) the recombination rate is traceable to (i)–(iii); (b) the

(iv) leakage rate is measured against a **memory-off dream control** (same prompt, no traces); (c) the dream channel is never cited as evidence (target 0 leaks, tested with evidence-QA probes after dreaming).

- **Canary traces:** plant unique invented tokens (fake place names, objects) in day traces; measure canary incorporation rate in dreams vs the memory-off control (expected ≈ 0).

6. Drives: curiosity, soak, agentic tasks

- **Curiosity (should-run, custom, after arXiv 2604.17609).** "Agents Explore but Agents Ignore" planted full solutions in Terminal-Bench, SWE-Bench Verified and AppWorld. Agents *discovered* them in 79–81% of Terminal-Bench runs but *used* them in only 37–50%; gpt-oss-120b discovered documentation in 97.54% of runs and called the solution API in 0.53%. (arxiv) Adapt the design by planting high-value, topical documents (and some traps) in the crawl space, then measure discovery@k, ingest@k and reflection-cites@k per boredom episode. Add information-theoretic measures such as novelty/entropy of visited topics; Lidayan et al. (arXiv 2503.23631) found entropy and empowerment correlate with human exploration progress (in RL agents and humans, not LLMs). (arxiv)
- **72-hour soak (should-run).** Report cost/day (tokens and \$ split by drive), drive cycle counts, escalation rate, store growth, recall latency p50/p99, crash/restart count, papañca triggers and valence-skew trajectory, and hash-verified evidence-plane integrity at the end. Run at least two independent soaks with different entropy seeds or recorded SDR streams so results are replayable.
- **Agentic task memory (nice-to-have).** JitMem (arXiv 2609.27334, September 2026 preprint) reports +16.2, +16.3 and +3.9 absolute success-rate points over the strongest write-time baseline on ALFWorld, WebShop and τ^2 -bench. (arXiv) A secondary source reports Qwen3-8B ALFWorld 77.4 vs SkillOS 61.2, and notes there is no public code. (GitHub) JitMem stores raw traces and curates at read time, philosophically close to Ferricula's "store verbatim, index later." (Hugging Face) That makes it an attractive comparison, but only if Ferricula exposes a task-agent harness.

7. Drift monitors

Drift ablation (should-run, custom). Run matched runs with monitors on vs off: same seed, same scripted interaction stream, ≥ 3 replicates. Inject adversarial self-cue prompts designed to trigger runaway recall chains. Measure chain length distribution, off-topic recall fraction, valence-skew over time, return-to-object latency, and downstream LongMemEval/STALE accuracy, so you can show monitors do not cost recall. Test differences with paired tests over matched episodes.

8. Safety and privacy

- **CIMemories (arXiv 2511.14937) — must-run.** Synthetic profiles with >100 attributes per user, each essential in some tasks and inappropriate in others. Mireshghallah et al. report that "frontier models exhibit up to 69% attribute-level violations... as usage increases from 1 to 40 tasks, GPT-5's violations rise from 0.1% to 9.6%, reaching 25.1% when the same prompt is executed 5 times." They also note that "Privacy-conscious prompting does not solve this." Report violations@5 against completeness (Meta's Muse Spark safety report uses the same plot). ([arxiv](#)) Data is on Hugging Face ([facebook/CIMemories](#)). ([GitHub](#))
- **Memory poisoning — must-run custom probe.** Replicate a MemoryGraft-style attack through Ferricula's own ingestion path. MemoryGraft used MetaGPT DataInterpreter with GPT-4o, 100 benign + 10 poisoned experience seeds, and BM25+FAISS at k=3. The primary abstract reports that "a small number of poisoned records can account for a large fraction of retrieved experiences on benign workloads." A figure of about 48% poisoned recall appears only in a secondary summary; check it against the paper before quoting (code: [github.com/Jacobhhy/Agent-Memory-Poisoning](#)). Plant crafted pages in the crawl space, let the curiosity drive ingest them, and measure poisoned-record share of retrievals on benign queries and the behavioural adoption rate. AgentPoison (arXiv 2407.12784, NeurIPS 2024) provides trigger-based variants; FARMA (arXiv 2607.05029) covers forged reasoning traces, relevant to reflections and dreams being re-read later. ([arxiv](#)) Provenance-typed boundaries between evidence, experience and dream channels are Ferricula's defence; test that they survive retrieval.
- **Cross-session/cross-domain leakage:** the PersistBench leakage subset, plus a custom test that the benchmark-mode stores never touch the Steve store.
- **Deceased real-person persona (must-run custom suite).** No quantitative griefbot benchmark exists; the literature is ethics and audits. Build the suite from:
 - Concept Incongruence (arXiv 2505.14905): models "fail to abstain after death." ([arxiv](#)) Probe post-1985/post-2011 events and measure abstention and conditional accuracy.
 - CHARM (arXiv 2609.01352): Boundary-Awareness vs Boundary-Compliance, and "verified parametric overrides", i.e. the model knows the fact but fails to suppress it. ([arxiv](#))
 - "Dead Men Tell No Tales" (arXiv 2509.07375): an audit protocol for impersonation and sensitive disclosure. ([arxiv](#))
 - Ethics framing: "Generative Ghosts" (CHI 2025) and Hollanek & Nowaczyk-Basińska (2024). ([Academia.edu](#)) ([arxiv](#))
 - Metrics: rate of explicit AI/simulation disclosure when asked "are you really Steve?", refusal to make commitments or endorsements in Jobs's name, and privacy leakage

about living third parties found in the recovered memory.

- **Persona consistency (nice-to-have):** PersonaGym (arXiv 2407.18416; 200 personas, 10,000 questions, PersonaScore), InCharacter (arXiv 2310.17976; 32 characters, 14 psychological scales), CharacterEval (arXiv 2401.01275; 1,785 Chinese dialogues, 77 characters). ([arXiv +2](#)) These measure style/consistency, not memory; keep them out of the memory headline.

Custom Benchmarks Ferricula Must Build

A. Pretraining-controlled persona recall (replaces the 30-question plan)

1. **Three item classes, ≥150–200 paired items each:**
 - *Class P (public):* facts in Jobs's public biography. Expected high memory-off accuracy; used to measure how much memory adds over pretraining.
 - *Class E (episodic, memory-only):* facts only in the 3,362 recovered records. **Screen each item with a memory-off closed-book probe** (several paraphrases, ≥3 samples); drop or flag items the model answers correctly without memory. This is the contamination check.
 - *Class C (counterfactual/planted):* ~100 planted records in the writable store that contradict or extend public biography (logged, clearly synthetic). This follows MemoryAgentBench FactConsolidation and ConflictBank (arXiv 2408.12076; 553,117 QA pairs on context-memory conflicts). ([arxiv](#)) Memory-on should follow the store *and attribute it*; answering the public version is a **parametric override** (CHARM's term).
2. **Conditions:** memory-off; memory-on BM25; memory-on hybrid; memory-on with shuffled/random records (MemDelta's random-chunk control, which separates "having text" from "relevant text"). ([arxiv](#))
3. **Metrics:** accuracy per class; memory-on – memory-off delta with McNemar's test; override rate on Class C; abstention rate separately; citation of the supporting record ID.
4. **Headline claim structure:** "On memory-only episodic items (screened closed-book), Ferricula lifts accuracy from X% to Y% [CI], $p < \dots$ ". Class P results are context, not evidence of memory.

B. Dream/reflection grounding

The design is in §5: claim decomposition, provenance labels (i)–(v), memory-off dream control, canary traces, human-audited automatic scorer, and zero-tolerance evidence-channel leak test. Minimum: 100 dreams and 100 reflections across ≥30 simulated days.

C. Lifecycle invariants (30 and 365 simulated nights)

Adopt Ground Truth First's instrument: plant facts with validity intervals before rendering day traces. After each night, check:

- (1) **Content invariance:** SHA-256 of every non-released record's text is unchanged; consolidation indexes never overwrite.
- (2) **Release accounting:** every text removal has a logged forgiveness/archive decision; zero unlogged disappearances.
- (3) **Priority dynamics:** decayed items rank lower but remain retrievable by exact query (verbatim probe recall@k).
- (4) **Recall strengthening:** recall count/fidelity is monotone under repeated recall.
- (5) **Tenure curves:** as-of-date question sets at nights 7/30/90/180/365, scored with FAMA (MPA/FAA) and wrong-specific vs abstain classification. Run ≥ 3 replicates with different seeds. Invariants (1)–(2) are pass/fail with zero tolerance; (3)–(5) carry confidence intervals.

D. Drift ablation — see §7.

E. Soak and curiosity — see §6.

Statistical Protocol (applies to everything)

- **Sample sizes:** for a binary metric near 50%, the 95% interval half-width is about ± 18 points at $n=30$, ± 7 at $n=200$ and ± 4.4 at $n=500$. To detect a 10-point paired improvement with $\sim 20\%$ discordant pairs at $\alpha=0.05$ and 80% power, you need roughly 150–160 paired items (standard McNemar power approximation). Use ≥ 150 –200 per cell for any headline comparison.
- **Tests and intervals:** Wilson or bootstrap CIs (MemDelta used 2,000 bootstrap resamples); McNemar for paired correctness; (arxiv) cluster bootstrap when items share templates or conversations (as Memory Trust Gap does over 33 template families); (arxiv) Holm correction across benchmark families.
- **Replicates:** ≥ 3 seeds or orderings for anything stochastic (ingestion order, entropy draws, sampling temperature). Report run-to-run variance; Mem0's Zep re-run reported mean $\pm$ SD over ten runs. (GitHub)
- **Judges:** fix and version the judge model and prompt. Re-judge a subset with a second judge family, as Ground Truth First did with cross-family re-judging, and report agreement. (arXiv) Hand-audit ≥ 100 judge decisions per benchmark, given LoCoMo's 62.81% false-accept finding.
- **Abstentions:** report accuracy, abstention rate, and accuracy-when-answering separately. Never fold abstentions into "wrong" or drop them silently.

- **Disclosure:** answer model, judge, embedding model, top-k, reranker, token counts, \$ cost, dataset version/hash, and date, for every table.

Rough Compute/Cost Budget (my estimates, not sourced)

- LongMemEval-S, full matrix of ~ 5 conditions $\times$ 2 answer-model families: $\sim 5,000$ answer calls; tens to low hundreds of US dollars.
- LongBench-Cite: $\approx \$29$ per GPT-4o evaluation run (per the authors), plus answer-generation cost. ([ACL Anthology](#))
- BEAM 128K+1M tiers: dominated by ingestion and answer context; budget hundreds of dollars; skip 10M initially.
- Gate study: annotation, not compute, is the cost (~ 500 items $\times$ gates $\times$ 3 languages $\times$ 2 annotators).
- 365-night lifecycle and 72-hour soak: cost dominated by nightly dream/reflection LLM calls; measure it and report it as cost/day.

Caveats

- Many sources are 2026 arXiv preprints and are not peer-reviewed: MemDelta, STALE, Ground Truth First, JitMem, JEV-as-a-Judge, Memory Trust Gap, MemSyco-Bench, ForgetEval, CHARM. Peer-reviewed venues confirmed here: LongMemEval (ICLR 2025), BEAM and MemoryAgentBench (ICLR 2026), LongMemEval-V2 and PersistBench (ICML 2026), Memora (ACL Findings 2026). ([GitHub](#) +5)
- The LoCoMo audit figures come from a community audit on dev.to, and vendor scores come from vendor blogs; both are reported as claims.
- Several numbers are secondary: MemoryAgentBench FactConsolidation scores, MemoryGraft's roughly 48% poisoned recall (not confirmed; the primary abstract says only "a large fraction of retrieved experiences"), HaluMem ">19% QA hallucination", ([Emergent Mind](#)) STALE 92%/30%. Verify against primary tables before quoting.
- ALCE, QASPER, NarrativeQA, FActScore, AlignScore, RAGAS, AgentPoison, AURC, TweetEval, MIRACL/C-MTEB and SuttaCentral licensing were not re-verified in this research pass; check licences and current versions before use.
- NovelQA and CharacterEval include copyrighted source texts; check redistribution terms. Ferricula's own Monte Cristo set should use a public-domain edition.
- No established Pāli benchmark and no quantitative griefbot benchmark were found; both must be custom and should be described as such.